Interim Report: Reverse Engineering and Analysis of PINN Model

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Project: Ocean Water Heat Flux Prediction

1 Introduction

Physics-Informed Neural Networks (PINNs) are a class of neural networks that incorporate physical laws into their learning process, allowing them to solve complex scientific problems such as partial differential equations and simulations of physical phenomena.

The objective of this task is to **reverse engineer a provided PINN model implemented in PyTorch**, understand its execution flow, and lay the foundation for future modifications. The ultimate goal is to restructure the code into modular components and prepare a combined dataset to test the model at a small scale.

At this stage, the focus has been on **model exploration, execution flow analysis, and preliminary dataset investigation**, while code restructuring and dataset preparation remain ongoing tasks.

2 Methodology

2.1 Model Exploration and Analysis

The provided PINN model, implemented using `nn.Module` inheritance in PyTorch, was carefully analyzed. Key functional components, including the forward pass and interdependencies between layers and functions, were identified to understand the execution flow. This analysis provides insight into areas suitable for future modularization and fine-grained control.

2.2 Flowchart Creation

Execution flow diagrams were created using **Mermaid** to visualize the flow of data and function calls within the model. Current diagrams represent key functional modules such as:

- training Pipeline Steps and Flow
- Inference and Analysis Flow

These diagrams are considered the primary **deliverables** for this stage.

2.3 Dataset Exploration (Preliminary)

Potential data sources were explored to collect the parameters required for testing the model. In particular:

- **Google Earth** datasets for relevant environmental data.
- **ECMWF datasets** accessed via **CSAPI**.

At this stage, the work focuses on understanding and planning data collection; actual dataset merging and preprocessing remain **ongoing tasks**.

2.4 Planned Tasks

The next steps include:

- **Code Restructuring:** Modularize the model code into semantic segments to facilitate easier testing and debugging.
- **Dataset Preparation:** Merge multiple datasets into a single structured dataset suitable for small-scale testing of the model.

3 Assets / Deliverables

The primary deliverables produced at this stage are the **Mermaid execution flow diagrams** of the PINN model. These diagrams illustrate the model's functional components and execution pathways, providing a clear visual reference for further code restructuring and testing.

4 Results / Conclusions

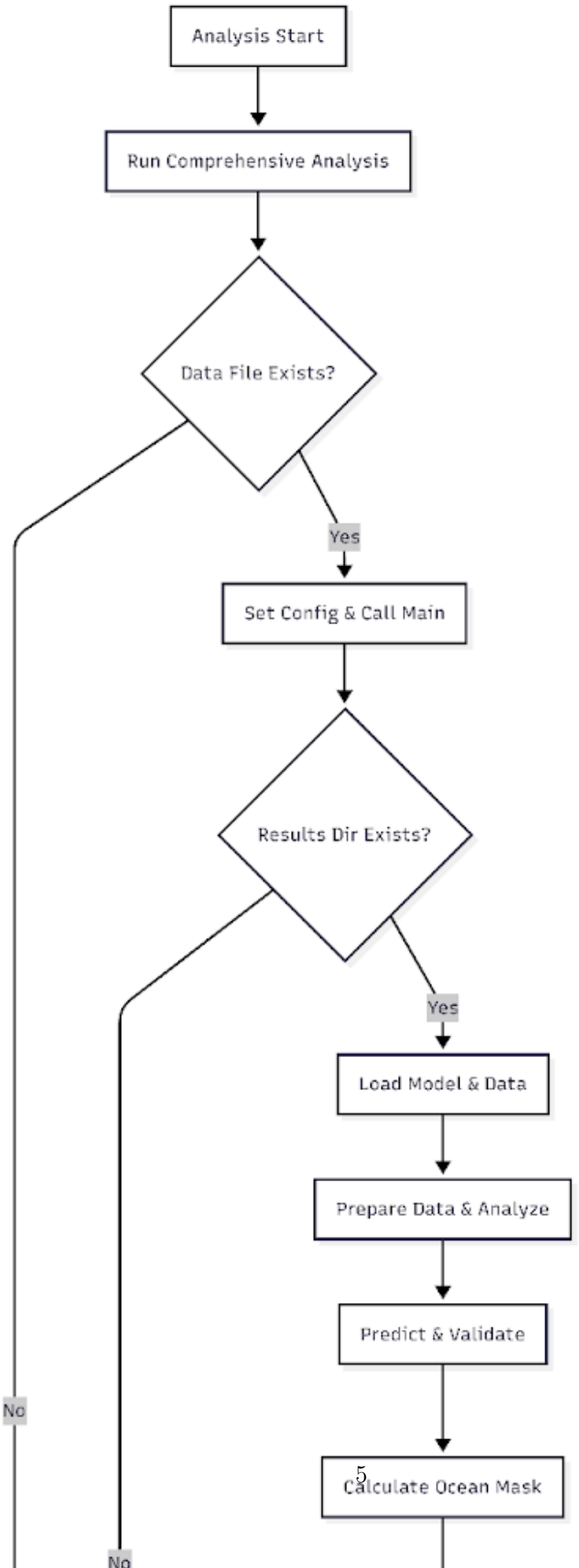
- Successfully generated **execution flow diagrams** to visualize data and function flow within the PINN model.
- Explored and analyzed the model's architecture to understand its dependencies and operational behavior.
- Investigated potential datasets from **Google Earth** and **ECMWF** to collect parameters required for testing the model.
- **Ongoing Work:**
  - Code restructuring for modularity and fine-grained control.
  - Dataset merging and preprocessing for small-scale model testing.

These preliminary tasks provide a solid foundation for the next phase, which will involve testing the model with a combined dataset and validating its execution flow.

5 Task Summary Table

Task	**Status**	**Notes**
Model exploration & analysis	Completed	PINN structure understood
Flowchart diagrams (Mermaid)	Completed	Execution flow captured
Dataset exploration	Ongoing	Google Earth & ECMWF explored via CSAPI
Code restructuring	Planned / Ongoing	Modularization next
Dataset merging	Planned / Ongoing	For small-scale testing
Presentation Research and Literature Survey	Started	Result Presentation

Table 1: Summary of Tasks Completed and Ongoing


Analysis Start
Run Comprehensive Analysis
Data File Exists?
Yes
No
Set Config & Call Main
Results Dir Exists?
Yes
No
Load Model & Data
Prepare Data & Analyze
Predict & Validate
Calculate Ocean Mask

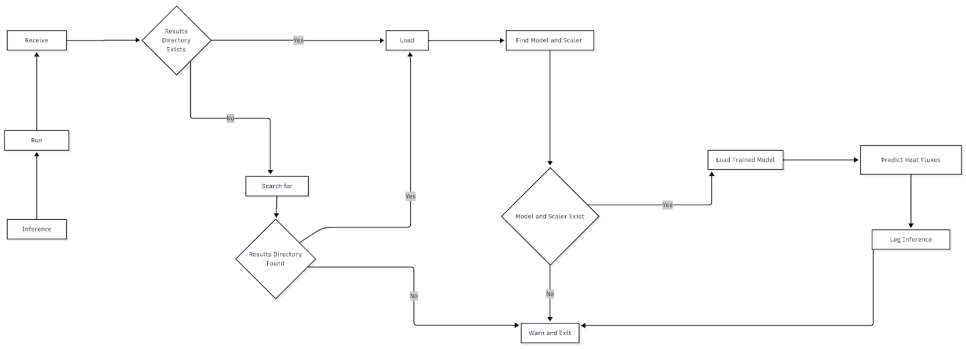


Figure 2: Default / Training Pipeline Steps

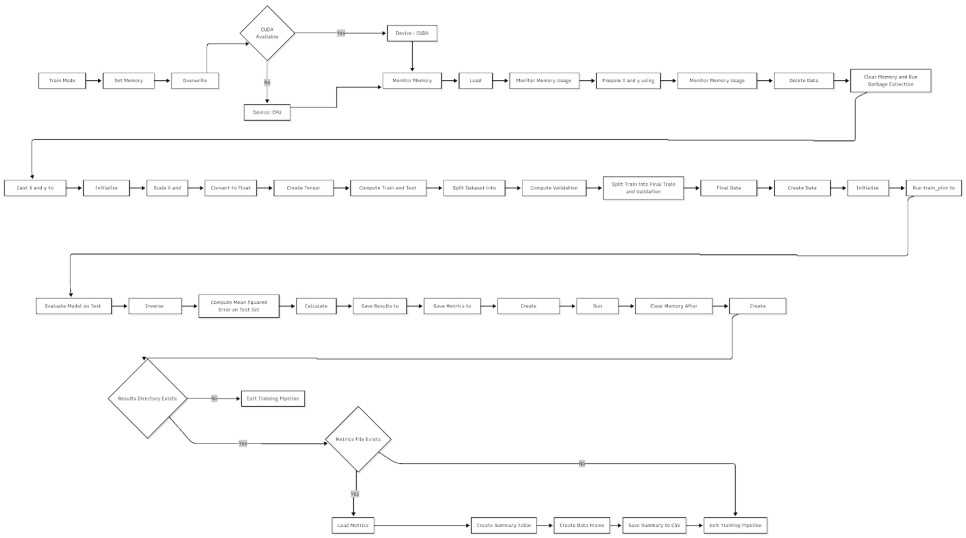


Figure 3: Inference Example Flow

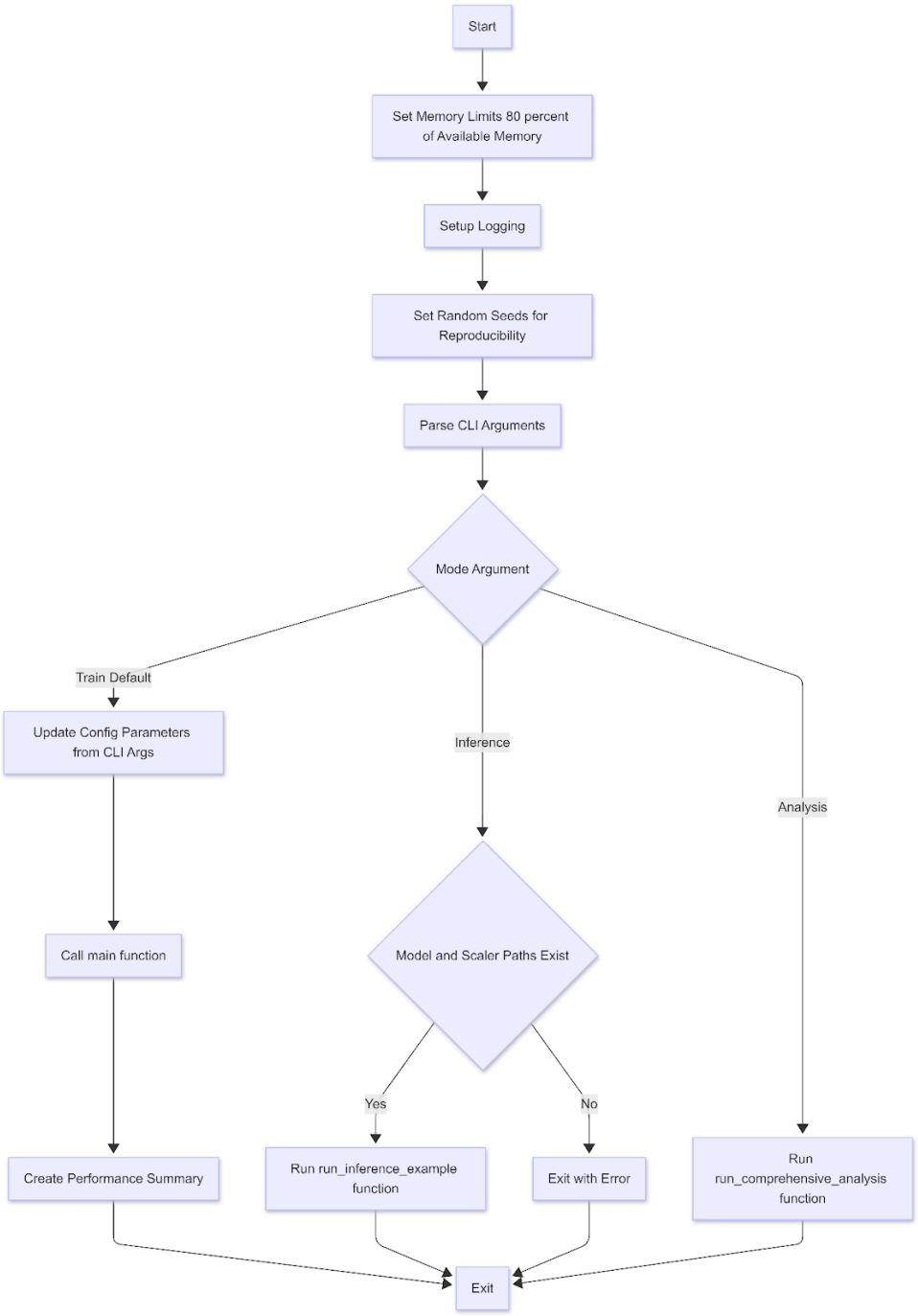


Figure 4: Analysis Steps and Flow